

Unsupervised Learning

Principal Components Analysis

Mahdi Shafiee kamalabad & Maarten Cruyff

1. Unsupervised Learning
2. Principal Component Analysis (PCA)
3. Application: Tumor Gene Expression Data

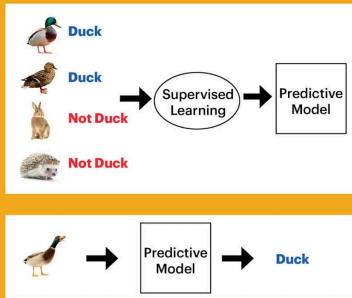
Supervised vs unsupervised learning

- **Supervised learning:** observations come with known labels
- **Unsupervised learning:** no labels, structure must be discovered (goal: clustering, dimension reduction (PCA))

Supervised vs unsupervised learning

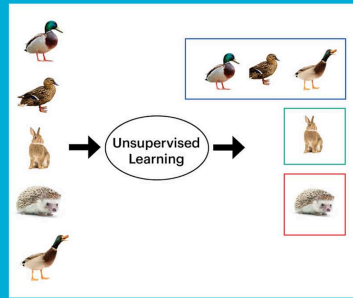
Known outcome

Supervised Learning (Classification Algorithm)

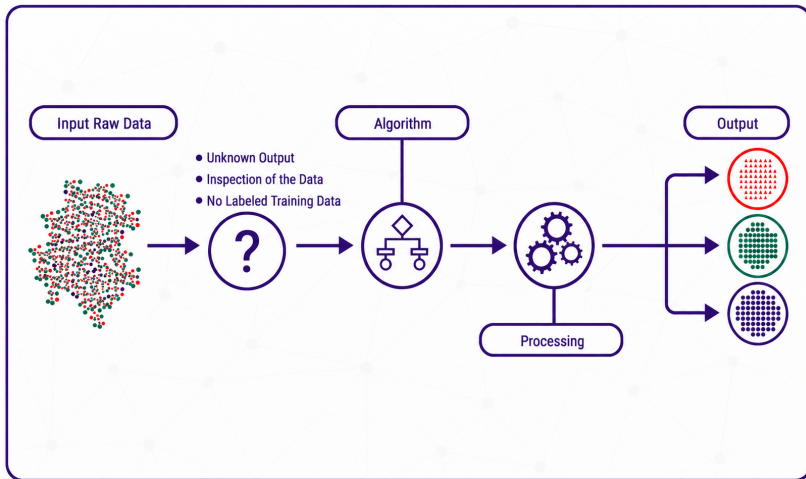


Unknown outcome

Unsupervised Learning (Clustering Algorithm)



Unsupervised learning



- **Clustering**
 - K-means
 - Hierarchical clustering
- **Dimension reduction**
 - Principal Component Analysis (PCA)
 - Multidimensional Scaling (MDS)
- **Density estimation**
 - Kernel density estimation
 - Gaussian mixture models

Principal Component Analysis (PCA)

Motivation: How can we summarize customer behavior?

- Customers are described by many variables
(e.g., *spending on fruits, vegetables, snacks, soft drinks, ...*)
- We want to reduce these variables to a few underlying dimensions
(e.g., *healthy vs. processed food preferences*)
- PCA helps us find the main directions of variation in the data



What is PCA?

PCA is a method for summarizing high-dimensional data using a small number of new variables called **principal components**.

These components capture the main patterns of variation in the data.

Principal Components

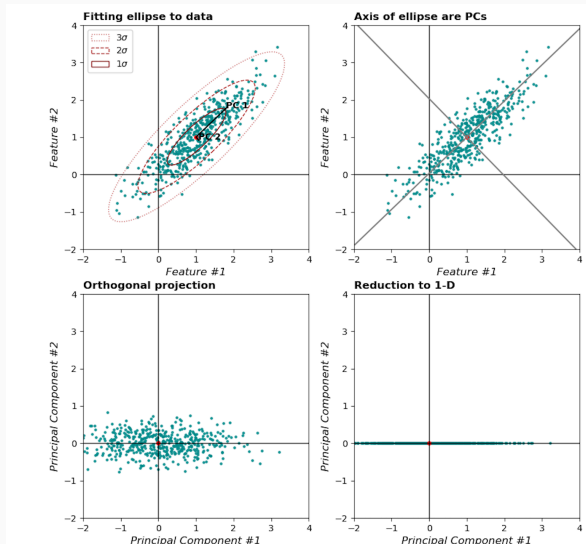
- New variables created from combinations of the original variables
- Orthogonal directions in the feature space
- Each component captures as much variance as possible
- Components are uncorrelated with each other

Dimension Reduction

- Keep the components that explain most of the variance
- Discard components that explain little variance
- Use fewer dimensions while preserving the main structure of the data

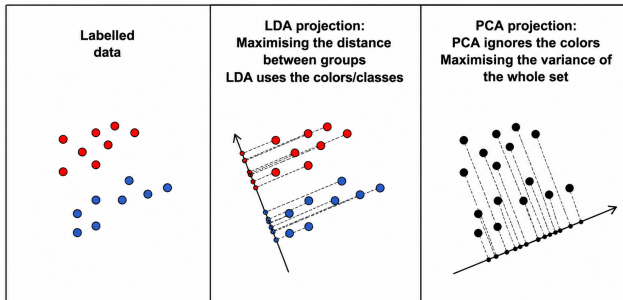
Dimension reduction in two-dimensional feature space

PCA finds and keeps the direction where the data varies the most, and removes less important directions.



PCA vs. LDA

- **PCA** (*unsupervised*):
 - Finds directions with maximum overall variance
 - Does not use class labels
- **LDA** (*supervised*):
 - Finds directions that best separate known groups
 - Uses class labels
 - Maximizes between-group separation relative to within-group variation



PCA of tumor data Implemented in R using the psych package

Tumor data

- Gene expression data
- each observation corresponds to a tumor sample
- each variable represents the expression level of a gene.
- Goal: to explore the dominant patterns of variation using PCA.

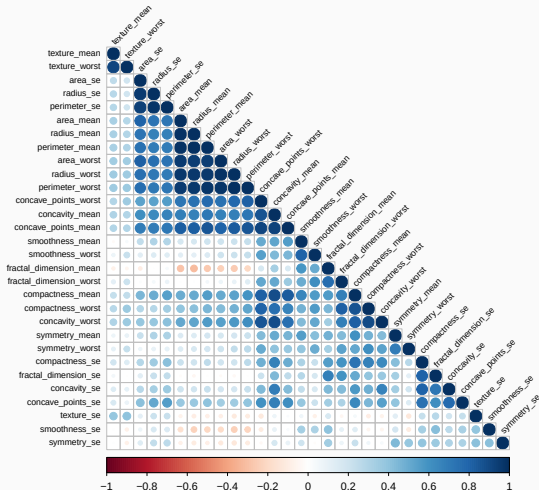
```
# A tibble: 4 x 31
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	M	18.0	10.4	123.	1001	0.118
2	M	20.6	17.8	133.	1326	0.0847
3	M	19.7	21.2	130	1203	0.110
4	M	11.4	20.4	77.6	386.	0.142

```
# i 25 more variables: compactness_mean <dbl>, concavity_mean <dbl>,  
# concave_points_mean <dbl>, symmetry_mean <dbl>,  
# fractal_dimension_mean <dbl>, radius_se <dbl>, texture_se <dbl>,  
# perimeter_se <dbl>, area_se <dbl>, smoothness_se <dbl>,  
# compactness_se <dbl>, concavity_se <dbl>, concave_points_se <dbl>,  
# symmetry_se <dbl>, fractal_dimension_se <dbl>, radius_worst <dbl>,  
# texture_worst <dbl>, perimeter_worst <dbl>, area_worst <dbl>, ...
```

PCA of tumor data

- PCA is most useful when variables are correlated
- Correlated genes may reflect shared biological processes

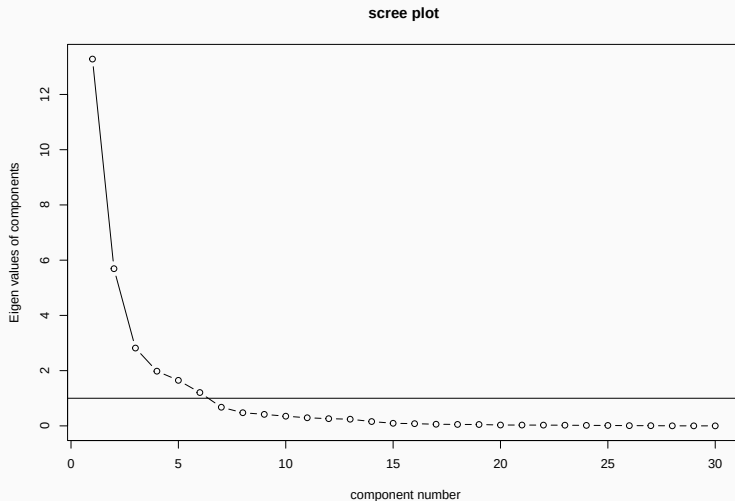


Step 2: How Many Components?

- The number of principal components equals the number of original features
(*here: 30*)
- Each principal component has an **eigenvalue**
 - The eigenvalue measures how much variance that component explains
 - The sum of all eigenvalues equals the total variance in the data
- A common rule of thumb:
 - Retain components with eigenvalue > 1
 - This is called the **Kaiser criterion**
 - It is mainly used when variables have been standardized
- Eigenvalues are visualized using a **scree plot**

Scree plot

- Displays eigenvalues in decreasing order
- Used to decide how many components to retain



Step 3: conduct the PCA

- Based on the scree plot, we retain 6 components

```
tumor_pca <- principal(tumor[, -1], nfactors = 6,  
                        rotate = "none")
```

- Proportion of variance explained by the 6 PC's

	PC1	PC2	PC3	PC4	PC5	PC6
SS loadings	13.282	5.691	2.818	1.981	1.649	1.207
Proportion Var	0.443	0.190	0.094	0.066	0.055	0.040
Cumulative Var	0.443	0.632	0.726	0.792	0.847	0.888
Proportion Explained	0.499	0.214	0.106	0.074	0.062	0.045
Cumulative Proportion	0.499	0.713	0.818	0.893	0.955	1.000

PCA Output: Key Rows

- **SS Loadings:** eigenvalues; variance explained by each component
- **Proportion Var:** percentage of total variance explained by each component
- **Cumulative Var:** total variance explained after adding components one by one
- **Proportion Explained:** percentage of variance explained among the selected components
- **Cumulative Proportion:** cumulative percentage among the selected components

	PC1	PC2	PC3	PC4	PC5	PC6
SS loadings	13.282	5.691	2.818	1.981	1.649	1.207
Proportion Var	0.443	0.190	0.094	0.066	0.055	0.040
Cumulative Var	0.443	0.632	0.726	0.792	0.847	0.888
Proportion Explained	0.499	0.214	0.106	0.074	0.062	0.045
Cumulative Proportion	0.499	0.713	0.818	0.893	0.955	1.000

Step 4: Interpreting the Loadings

The j -th principal component Z_j is a linear combination of the original variables:

$$Z_j = \phi_{1j}X_1 + \cdots + \phi_{pj}X_p, \quad j = 1, \dots, p$$

- ϕ_{ij} is the loading of variable X_i on component Z_j
- For standardized variables, loadings can be interpreted similarly to correlations
- Loadings indicate which variables contribute most to a component
- Common rule of thumb: $|\text{loading}| > 0.4$ indicates a strong contribution

Loadings

Loadings:

	PC1	PC2	PC3	PC4	PC5	PC6
radius_mean	0.798	-0.558				
perimeter_mean	0.829	-0.513				
area_mean	0.805	-0.551				
smoothness_mean	0.520	0.444			0.469	
compactness_mean	0.872					
concavity_mean	0.942					
concave_points_mean	0.951					
symmetry_mean	0.504	0.454				
radius_se	0.751		0.451			
perimeter_se	0.770		0.448			
area_se	0.739					
compactness_se	0.621	0.555				
concavity_se	0.560	0.470			-0.455	
concave_points_se	0.668					
radius_worst	0.831	-0.525				
perimeter_worst	0.862	-0.477				
area_worst	0.820	-0.523				
compactness_worst	0.766					
concavity_worst	0.834					
concave_points_worst	0.914					
fractal_dimension_mean		0.875				
fractal_dimension_se		0.668				
fractal_dimension_worst	0.480	0.657				
texture_se			0.629	0.506		
smoothness_se		0.488	0.518			
texture_mean				0.849		
texture_worst				0.891		
symmetry_se		0.439	0.484		-0.539	
symmetry_worst	0.448		-0.455		-0.548	
smoothness_worst	0.466	0.411	-0.436		0.417	0.406

Rotation

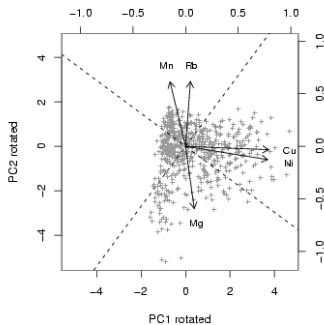
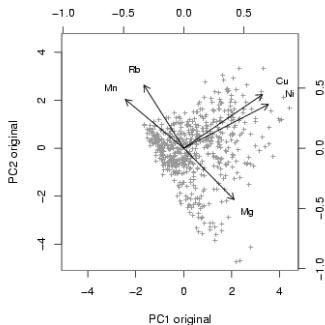
Rotation is used to make the components easier to interpret.

Before rotation, a variable may load moderately on several components.

After rotation, we try to get a simpler pattern:

- high loading on one component
- low loadings on the others

Rotation may redistribute variance across components, but it does not change the total variance explained by the retained components.



Remark: Rotation Options

Rotation can be used after selecting the number of components to make the loading pattern easier to interpret.

- **Orthogonal rotations**

Components remain uncorrelated.

- **Varimax**: most common; gives a simpler loading pattern
- **Quartimax**: emphasizes a general component
- **Equamax**: compromise between varimax and quartimax

- **Oblique rotations**

Components are allowed to be correlated.

- **Oblimin**
- **Promax**

Oblique rotations can be more realistic when the underlying concepts are related. For example, anxiety and depression may form different factors but still be correlated.

- In PCA, if rotation is used, **varimax** is commonly used

Rotated solution

- Based on the scree plot, we retain 6 components
- Apply varimax rotation to improve interpretability

```
tumor_pca <- principal(tumor[, -1], nfactors = 6,  
                        rotate = "varimax")
```

- Slightly different proportion of variance explained
- Rotation redistributes variance across components
- Total variance explained by the 6 components remains the same

	RC1	RC2	RC5	RC3	RC4	RC6
SS loadings	10.286	5.360	4.178	2.465	2.301	2.038
Proportion Var	0.343	0.179	0.139	0.082	0.077	0.068
Cumulative Var	0.343	0.522	0.661	0.743	0.820	0.888
Proportion Explained	0.386	0.201	0.157	0.093	0.086	0.077
Cumulative Proportion	0.386	0.588	0.744	0.837	0.923	1.000

loadings for the rotated PCA solution

- Focus on variables with large absolute loadings (rule of thumb: $|\text{loading}| > 0.4$)

Loadings

```
print(loadings(tumor_pca), sort = T, cex = c(1, 1), cutoff = .4)
```

Loadings:

	RC1	RC2	RC5	RC3	RC4	RC6
radius_mean	0.955					
perimeter_mean	0.955					
area_mean	0.970					
concavity_mean	0.662	0.565				
concave_points_mean	0.815					
radius_se	0.827					
perimeter_se	0.815					
area_se	0.872					
radius_worst	0.952					
perimeter_worst	0.948					
area_worst	0.954					
concave_points_worst	0.700		0.458			
compactness_mean	0.475	0.569	0.543			
compactness_se		0.896				
concavity_se		0.887				
concave_points_se		0.732				
fractal_dimension_se		0.862				
concavity_worst	0.459	0.575	0.432			
smoothness_mean			0.833			
fractal_dimension_mean		0.568	0.589			
smoothness_worst			0.943			
compactness_worst		0.529	0.539			
fractal_dimension_worst		0.564	0.658			
texture_se				0.640	0.522	
smoothness_se				0.768		
texture_mean					0.909	
texture_worst					0.953	
symmetry_mean			0.418			0.721
symmetry_se				0.410		0.769

Component Scores

The component score Z_{ij} for observation i on component j is computed as:

$$Z_{ij} = \phi_{1j}X_{i1} + \cdots + \phi_{pj}X_{ip}, \quad i = 1, \dots, n$$

where ϕ_{kj} is the loading/weight of variable k on component j .

Component scores summarize where each observation lies on a principal component.

Relationship Between Loadings and Scores

A **biplot** visualizes both parts of a PCA solution:

- **Component scores:** show the observations
 - Points represent observations using their component scores
- **Component loadings:** show the variables
 - Arrows represent variables using their component loadings

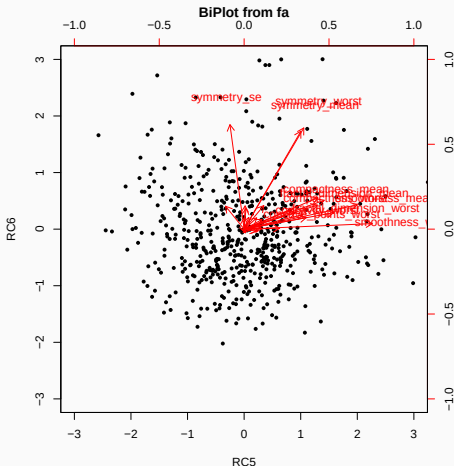
In the next slide, we show a biplot of **RC3** and **RC6**:

- **RC3** mainly reflects the `smoothness_se` component
- **RC6** mainly reflects the `symmetry` component

Focused Biplot: RC3 vs. RC6

The full biplot matrix can be crowded, so we focus on one pair of components at a time.

- **RC5** mainly reflects the `smoothness_se` component
- **RC6** mainly reflects the `symmetry` component



Interpreting the RC3–RC6 Biplot

This biplot shows both tumors/samples and variables:

- **Black points** represent tumors/samples
- **Red arrows** represent variables

The horizontal axis is **RC5**, which mainly reflects the `smoothness_se` component. The vertical axis is **RC6**, which mainly reflects the `symmetry` component.

Most black points are clustered near the center, meaning most tumors have fairly average scores on **RC5** and **RC6**.

The red arrows show which variables drive these two components:

- Arrows pointing to the **right** are more related to **RC5**
 - For example, `smoothness_se` and `texture_se` point mostly right
 - Tumors far to the right tend to have higher values on these variables
- Arrows pointing **upward** are more related to **RC6**
 - Several symmetry-related variables point upward
 - Tumors higher in the plot tend to have higher symmetry-related values
- Tumors far from the center are more unusual or extreme on these component patterns

Why Use a Biplot?

A biplot helps us interpret PCA results in terms of the original variables.
This helps us:

- identify unusual tumors
- see groups of similar tumors
- understand which variables drive the differences
- explain components using the original measurements

Summary

PCA reduced the dimensionality of the data:

- from 30 original features to 6 components
- while retaining 88.8% of the total variance

Advantages

- **Improves understanding of the data**
 - component loadings show which features vary together
- **Simplifies representation of observations**
 - component scores summarize tumors in a low-dimensional space
 - scores can be used in later analyses, such as classification or prediction
- **Can reduce noise**
 - low-variance components are often less informative and may be removed

Overview: PCA Functions in R: psych package

```
# Scree plot using the correlation matrix
VSS.scree(cor(tumor[, -1]))

# PCA with 6 components and varimax rotation
tumor_pca <- principal(
  tumor[, -1],
  nfactors = 6,
  rotate = "varimax",
  scores = TRUE
)

# Variance explained by the components
tumor_pca$Vaccounted

# Component loadings
print(loadings(tumor_pca), cutoff = 0.4, sort = TRUE)

# Component scores
head(tumor_pca$scores)

# Biplot matrix for rotated components
psych::biPlot(tumor_pca, choose = c(3, 4), cut1 = 0.4)
```


Overview: PCA Functions in R: Base R

```
# PCA using standardized variables
x <- prcomp(tumor[, -1], center = TRUE, scale. = TRUE)

# Scree plot
screeplot(x, npcs = 6)
screeplot(x, npcs = 6, type = "lines") # line

# One biplot: PC1 vs PC2
biplot(x)

# Specific pair: PC3 vs PC6
biplot(x, choices = c(3, 6))
```

Applications of PCA

1. Dimension reduction

- retain most of the variance using fewer components
- use of component scores as predictors in supervised models for prediction

2. Scale construction

- personality traits (e.g. Big Five)
- psychological and aptitude tests (career choice, IQ)

3. Text analysis

- exploratory analysis of writing styles
- authorship attribution (e.g. Dutch national anthem)

4. Pattern recognition

- image analysis (e.g. eigenfaces)

5. and many more Lab 4a